

# ASSESSMENT TOOL 1 – Application-Based Questions

## ITA0609 - Machine Learning

Sameer Ahmed G

192421154

### Question 1

#### Problem Statement

A parking facility has three sections containing cars of different colors.

- **Section A:** 5 Red, 3 Blue, 2 Black cars
- **Section B:** 4 Red, 6 Blue, 5 Black cars
- **Section C:** 7 Red, 2 Blue, 6 Black cars

A section is chosen at random, but **Section B is twice as likely** to be selected as each of the other sections due to higher usage. A randomly selected car from the chosen section turns out to be **blue**.

Determine the probability that the selected blue car came from **Section C**.

#### Introduction

This problem is an application of **Bayes' Theorem**, which is widely used in Machine Learning, Artificial Intelligence, Medical Diagnosis, Spam Filtering, and Decision-Making Systems.

Bayes' Theorem helps us determine the probability of an event after observing new evidence. In this problem, the evidence is that the selected car is blue, and we need to determine the probability that it came from Section C.

#### Given Data

##### Number of Cars

| Section | Red | Blue | Black | Total |
|---------|-----|------|-------|-------|
| A       | 5   | 3    | 2     | 10    |
| B       | 4   | 6    | 5     | 15    |
| C       | 7   | 2    | 6     | 15    |

Since Section B is twice as likely to be selected,

Let

$$\begin{aligned}P(A) &= x \\P(B) &= 2x \\P(C) &= x\end{aligned}$$

Since probabilities add to 1,

$$\begin{aligned}x + 2x + x &= 1 \\4x &= 1 \\x &= \frac{1}{4}\end{aligned}$$

Therefore,

$$\begin{aligned}P(A) &= \frac{1}{4} \\P(B) &= \frac{1}{2} \\P(C) &= \frac{1}{4}\end{aligned}$$

### **Step 1: Probability of Selecting a Blue Car from Each Section**

#### **Section A**

Total cars = 10

Blue cars = 3

$$P(\text{Blue} \mid A) = \frac{3}{10} = 0.3$$

#### **Section B**

Total cars = 15

Blue cars = 6

$$P(\text{Blue} \mid B) = \frac{6}{15} = 0.4$$

### Section C

Total cars = 15

Blue cars = 2

$$P(\text{Blue} \mid C) = \frac{2}{15} = 0.1333$$

### Step 2: Calculate the Total Probability of Selecting a Blue Car

Using the Law of Total Probability,

$$P(\text{Blue}) = P(A)P(\text{Blue} \mid A) + P(B)P(\text{Blue} \mid B) + P(C)P(\text{Blue} \mid C)$$

Substituting the values,

$$\begin{aligned} &= \left(\frac{1}{4} \times \frac{3}{10}\right) + \left(\frac{1}{2} \times \frac{6}{15}\right) + \left(\frac{1}{4} \times \frac{2}{15}\right) \\ &= 0.075 + 0.20 + 0.0333 \\ P(\text{Blue}) &= 0.3083 \end{aligned}$$

### Step 3: Apply Bayes' Theorem

Bayes' Theorem is

$$P(C \mid \text{Blue}) = \frac{P(C) \times P(\text{Blue} \mid C)}{P(\text{Blue})}$$

Substitute the values,

$$\begin{aligned} &= \frac{\frac{1}{4} \times \frac{2}{15}}{0.3083} \\ &= \frac{0.0333}{0.3083} \\ &= 0.1081 \end{aligned}$$

### Final Answer

$$P(C \mid \text{Blue}) = 0.1081$$

Therefore,

$$\boxed{P(C \mid \text{Blue}) \approx 10.8\%}$$

## Interpretation

The probability that the selected blue car belongs to **Section C** is approximately **10.8%**.

Although the car is blue, Section C has only **2 blue cars out of 15**, whereas Section B has **6 blue cars out of 15** and is also selected more frequently. Therefore, the probability of the blue car coming from Section C is relatively low.

## Application in Machine Learning

Bayes' Theorem is extensively used in Machine Learning for:

- Spam Email Detection
- Medical Diagnosis
- Fraud Detection
- Weather Prediction
- Recommendation Systems
- Naïve Bayes Classification

It enables the model to update probabilities whenever new evidence becomes available.

## Conclusion

Using Bayes' Theorem, the probability that the randomly selected blue car came from Section C is **10.8%**. This demonstrates how conditional probability helps in making informed predictions based on observed evidence.

## Question 2

### Problem Statement

The training and validation loss of a machine learning model are recorded over eight epochs.

| Epoch | Training Loss | Validation Loss |
|-------|---------------|-----------------|
| 1     | 2.8           | 2.4             |
| 2     | 2.5           | 2.2             |
| 3     | 2.2           | 2.0             |
| 4     | 2.0           | 2.1             |
| 5     | 1.8           | 2.3             |
| 6     | 1.6           | 2.6             |
| 7     | 1.5           | 2.9             |
| 8     | 1.3           | 3.2             |

Answer the following:

1. Identify the epoch at which overfitting begins.
2. Suggest two methods to reduce overfitting.
3. Explain how overfitting affects generalization.

### Introduction

Training a machine learning model involves minimizing the training loss while ensuring that the model performs well on unseen data.

Two important metrics are:

- **Training Loss:** Error on training data.
- **Validation Loss:** Error on unseen validation data.

A good model should have both losses decreasing together. If training loss continues to decrease while validation loss increases, the model is said to be **overfitting**.

### Observation of Loss Values

| Epoch | Training Loss | Validation Loss | Observation                 |
|-------|---------------|-----------------|-----------------------------|
| 1     | 2.8           | 2.4             | Both losses decreasing      |
| 2     | 2.5           | 2.2             | Model learning              |
| 3     | 2.2           | 2.0             | Best validation performance |

|   |     |     |                           |
|---|-----|-----|---------------------------|
| 4 | 2.0 | 2.1 | Validation loss increases |
| 5 | 1.8 | 2.3 | Overfitting increases     |
| 6 | 1.6 | 2.6 | Strong overfitting        |
| 7 | 1.5 | 2.9 | Poor generalization       |
| 8 | 1.3 | 3.2 | Severe overfitting        |

### i) Epoch Where Overfitting Begins

Validation loss reaches its minimum at **Epoch 3**.

From **Epoch 4**, validation loss starts increasing while training loss continues decreasing.

This indicates that the model is learning unnecessary details and noise from the training data.

Therefore,

**The model starts overfitting at Epoch 4.**

### ii) Techniques to Reduce Overfitting

#### 1. Early Stopping

Early stopping monitors validation loss during training.

When validation loss begins increasing, training is stopped.

In this example, the model should stop after **Epoch 3**.

Advantages:

- Prevents memorization
- Saves training time
- Improves prediction accuracy

#### 2. Regularization

Regularization reduces model complexity.

Popular techniques include:

- L1 Regularization
- L2 Regularization
- Dropout

These methods penalize large weights and prevent the model from fitting noise.

Advantages:

- Better generalization
- Lower validation loss
- Improved robustness

### iii) Effect of Overfitting on Generalization

Generalization is the ability of a model to perform well on unseen data.

An overfitted model:

- Memorizes training data.
- Learns noise instead of meaningful patterns.
- Performs exceptionally well on training data.
- Performs poorly on new data.
- Produces inaccurate predictions in real-world scenarios.

Hence, overfitting significantly reduces the model's generalization capability.

### Applications

Avoiding overfitting is important in:

- Image Classification
- Speech Recognition
- Medical Diagnosis
- Financial Forecasting
- Self-Driving Cars
- Face Recognition

### Conclusion

The model begins overfitting at **Epoch 4** because validation loss increases while training loss decreases. Techniques such as **early stopping** and **regularization** help reduce overfitting and improve the model's ability to generalize to unseen data.

### Question 3

#### Problem Statement

The training and validation accuracy of a machine learning model are recorded over eight epochs.

| Epoch | Training Accuracy | Validation Accuracy |
|-------|-------------------|---------------------|
| 1     | 60%               | 58%                 |
| 2     | 65%               | 63%                 |
| 3     | 70%               | 68%                 |
| 4     | 75%               | 72%                 |
| 5     | 80%               | 73%                 |
| 6     | 85%               | 72%                 |
| 7     | 88%               | 70%                 |
| 8     | 92%               | 68%                 |

Answer the following:

1. Identify the epoch where overfitting starts.
2. Explain why validation accuracy decreases while training accuracy increases.
3. Suggest two methods to improve validation performance.

#### Introduction

Accuracy measures how many predictions made by a model are correct.

Two important measures are:

- **Training Accuracy:** Accuracy on training data.
- **Validation Accuracy:** Accuracy on unseen data.

An ideal model shows improvements in both training and validation accuracy. If training accuracy continues increasing while validation accuracy decreases, it indicates overfitting.

#### Observation

| Epoch | Training Accuracy | Validation Accuracy | Observation     |
|-------|-------------------|---------------------|-----------------|
| 1     | 60%               | 58%                 | Learning begins |
| 2     | 65%               | 63%                 | Model improves  |
| 3     | 70%               | 68%                 | Better learning |



|   |     |     |                             |
|---|-----|-----|-----------------------------|
| 4 | 75% | 72% | Good performance            |
| 5 | 80% | 73% | Highest validation accuracy |
| 6 | 85% | 72% | Validation decreases        |
| 7 | 88% | 70% | Overfitting increases       |
| 8 | 92% | 68% | Severe overfitting          |

### i) Identify the Epoch Where Overfitting Starts

Validation accuracy reaches its highest value at **Epoch 5 (73%)**.

After this point, validation accuracy decreases while training accuracy continues increasing.

Therefore,

**The model starts overfitting at Epoch 6.**

### ii) Why Does Validation Accuracy Decrease?

The model begins memorizing the training data instead of learning meaningful features.

As a result:

- Training accuracy increases because the model performs better on familiar data.
- Validation accuracy decreases because the model cannot correctly classify unseen data.
- The model becomes too specialized and loses its ability to generalize.

This phenomenon is known as **overfitting**.

### iii) Methods to Improve Validation Performance

#### 1. Early Stopping

Stop training when validation accuracy reaches its maximum (Epoch 5).

Benefits:

- Prevents overfitting
- Improves prediction accuracy
- Saves computation time

## 2. Data Augmentation

Generate additional training data by applying transformations such as:

- Rotation
- Flipping
- Scaling
- Cropping
- Noise addition

Benefits:

- Increases dataset diversity
- Reduces overfitting
- Improves validation accuracy

*(Alternatively, you can mention regularization or dropout as acceptable methods.)*

## Applications

Monitoring training and validation accuracy is essential in:

- Medical Image Analysis
- Face Recognition
- Autonomous Vehicles
- Fraud Detection
- Sentiment Analysis
- Recommendation Systems

## Conclusion

The model begins overfitting at **Epoch 6**, as validation accuracy decreases while training accuracy continues to improve. This indicates that the model is memorizing the training data rather than learning general patterns. Techniques such as **early stopping** and **data augmentation** improve validation performance and help the model generalize better to unseen data.